

Physiology: Basic Concepts in Respiratory Physiology

PPOM 2 - Respiratory System

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Associate Professor

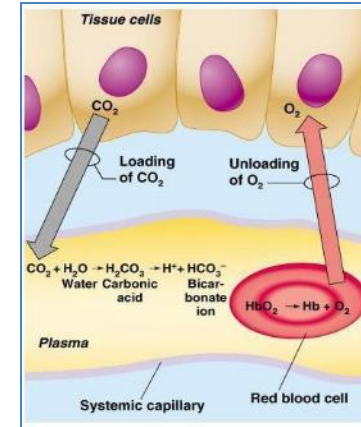
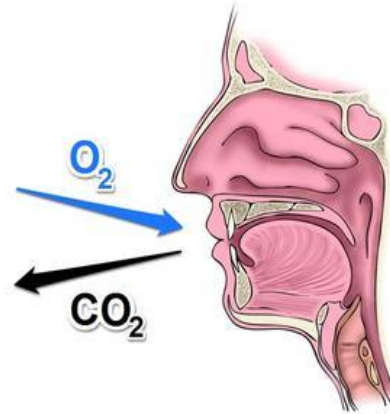
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Learning Objectives

1. Compare and contrast the relationships between lung volumes and capacities.
2. Discuss spirometry and its limitations in measurements of lung volumes and capacities.
3. Outline methods to determine residual volume and functional residual capacity.
4. Extrapolate lung volumes and capacities with respect to obstructive and restrictive lung diseases.
5. Demarcate regions of the lung that do not participate in gas exchange and learn methods to measure dead space.

Functions of Respiratory System



Gaseous exchange between the atmosphere, blood and body tissue cells

Protection against inhaled substances and pathogens

Acid-Base Homeostasis

Vocal Control

Four Processes

- Ventilation
- Diffusion
- Transport
- Regulation

Basic Concepts in Respiratory Physiology

1. Lung Volumes and Capacities
2. Pulmonary Test : Spirometry
3. Lungs Conditions: Restrictive and Obstructive Disease
4. Dead Space

Pulmonary Function Tests (PFTs)

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Pulmonary Function Lab

Date: _____

Name: _____ ID: _____ Room: _____

Gender: _____ Height(cm): _____ Weight(kg): _____ Age: _____ Body Mass Index: _____

Medication: _____ Physician: _____ Technician: _____ Smoker: _____ How Long: _____

Quit: _____ Stopped: _____ yrs

			#7	#8	#9	#10	#11	#12	#13
			Ref	CI	Pre	% Ref	Post	% Ref	% Chg
Spirometry									
#1	FVC	Liters	3.19	(2.5 - 3.9)	(2.33)	(73)	(2.30)	(72)	-1
#2	FEV1	Liters	2.62	(2.1 - 3.2)	(1.01)	(39)	(1.01)	(38)	-0
#3	FEV1/FVC	%	82	(72.6 - 90.7)	(43)		(44)		
	FEV1/SVC	%			40				
	FEF25-75%	L/sec	2.85	(1.5 - 4.2)	(0.40)	(14)	(0.38)	(13)	-6
	FEF50%	L/sec	3.50	(1.7 - 5.3)	(0.46)	(13)	(0.39)	(11)	-16
	PEF	L/sec	5.85	(3.0 - 8.7)	3.86	66	3.68	63	-5
	FET100%	Sec			9.61		11.08		15
	FIF50%	L/sec	5.50		2.14	39	2.68	49	26
	FEF/FIF50				0.22		0.14		-33
	FVL ECode				000010		000010		
Lung Volumes									
#4	TLC	Liters	4.90	(3.8 - 6.0)	(5.99)	(122)			
	VC	Liters	3.19	(2.5 - 3.9)	2.55	80			
	IC	Liters	2.10	(1.7 - 2.5)	1.87	89			
	FRC PL	Liters	2.73	(1.7 - 3.8)	(4.12)	(151)			
	ERV	Liters	1.00		0.62	62			
#5	RV	Liters	1.74	(1.0 - 2.5)	(3.43)	(198)			
	RV/TLC	%	35	(24.2 - 46.2)	(57)				
	LVol ECode				000002				
Diffusing Capacity									
#6	DLCO	mL/mmHg/min	22.3	(15.8 - 28.8)	16.4	74			
	DL Adj	mL/mmHg/min	22.3	(15.8 - 28.8)	16.4	74			
	DLCO/VA	mL/mHg/min/L	5.33	(4.0 - 6.6)	(3.75)	(70)			
	DLCO ECode				0000				
	TLC Sb	Liters	4.90	(3.8 - 6.0)	4.37	89			

#14

Comments: TESTS ARE PRE AND POST SALBUTAMOL 4 PUFFS.

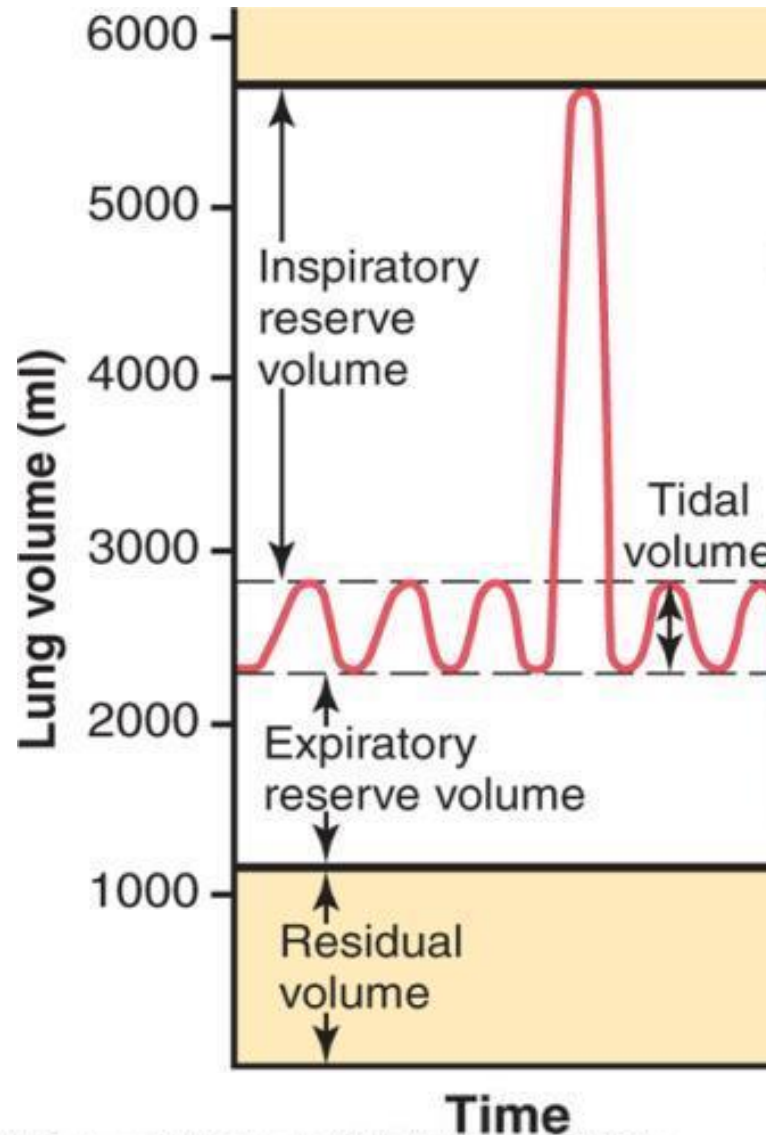
Flow

Volume

Time

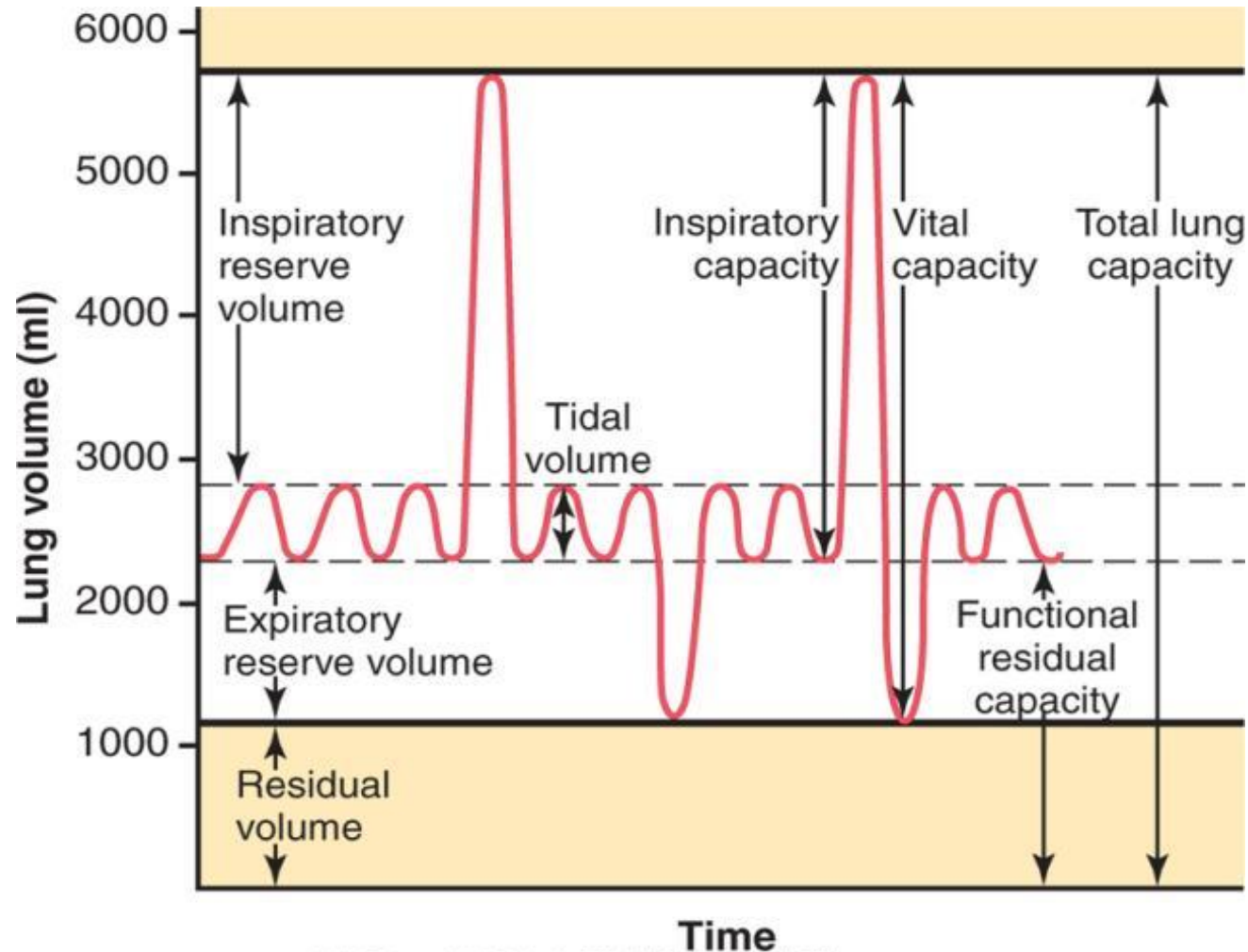
Page 1

Lung Volumes

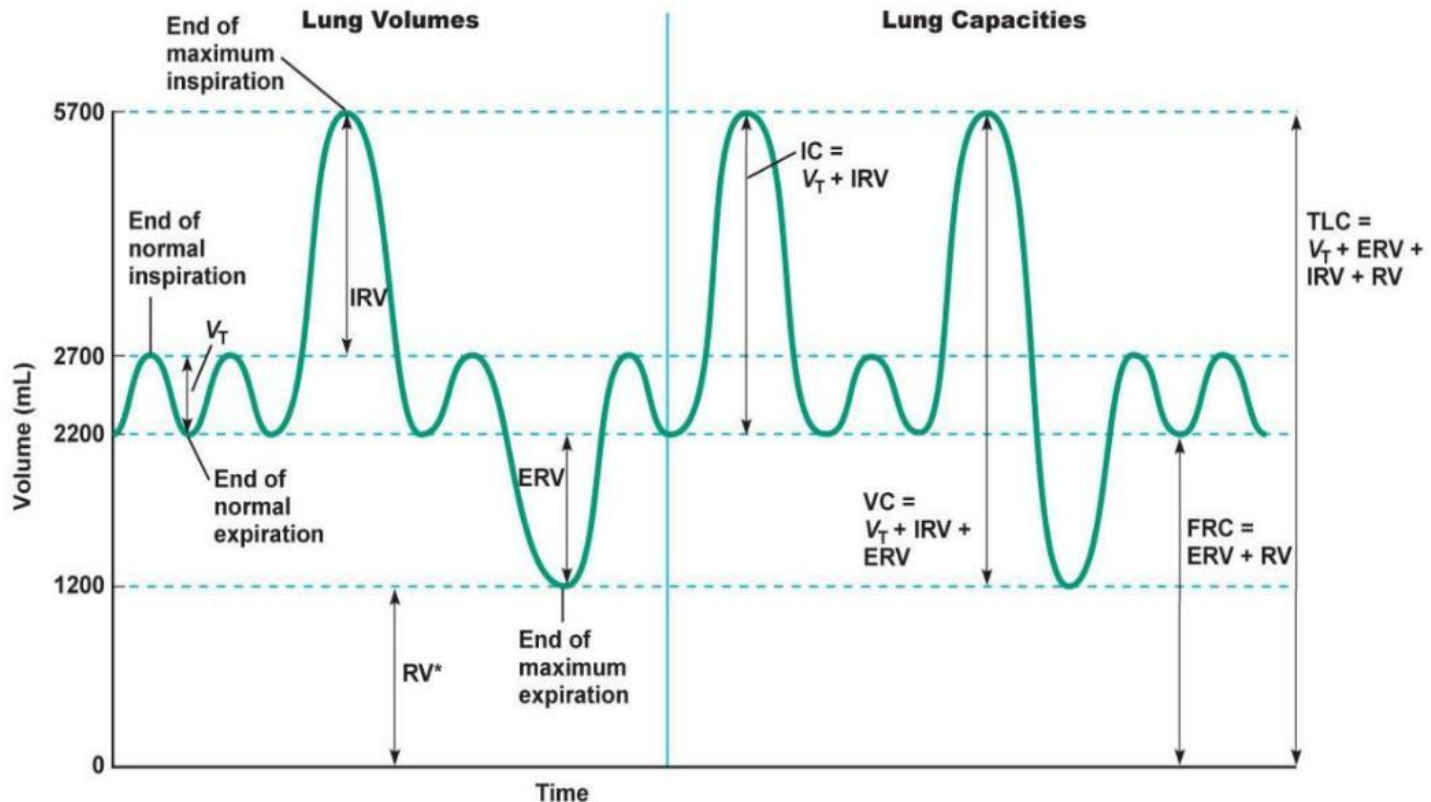


RV: Air that cannot
be blown out no
matter how hard
you try

Lung Capacities



Lung Capacities



Normal lung volumes and capacities for a healthy 70-kg male

Lung Volumes

V_T = Tidal volume = 500 mL
 IRV = Inspiratory reserve volume = 3000 mL
 ERV = Expiratory reserve volume = 1000 mL
 RV = Residual volume* = 1200 mL

*Cannot be measured by spirometry

Lung Capacities

IC = Inspiratory capacity = $V_T + IRV$ = 3500 mL
 VC = Vital capacity = $V_T + IRV + ERV$ = 4500 mL
 FRC = Functional residual capacity = $ERV + RV$ = 2200 mL
 TLC = Total lung capacity = $V_T + ERV + IRV + RV$ = 5700 mL

Lung Capacities

IC: $VT + IRV$

The maximum amount of air that can be inspired following a normal expiration

VC: $VT + IRV + ERV$ or Forced Vital Capacity (FVC)

The maximum amount of air that can be expired following a maximal Inspiration

Most you can exhale

FRC: $RV + ERV$

The amount of air remaining in the lungs following a normal expiration.

Volume when the system is relaxed.

Chest wall pulling out = lungs pulling in

TLC: $RV + ERV + IRV + VT$ Sum of all volumes

Summary

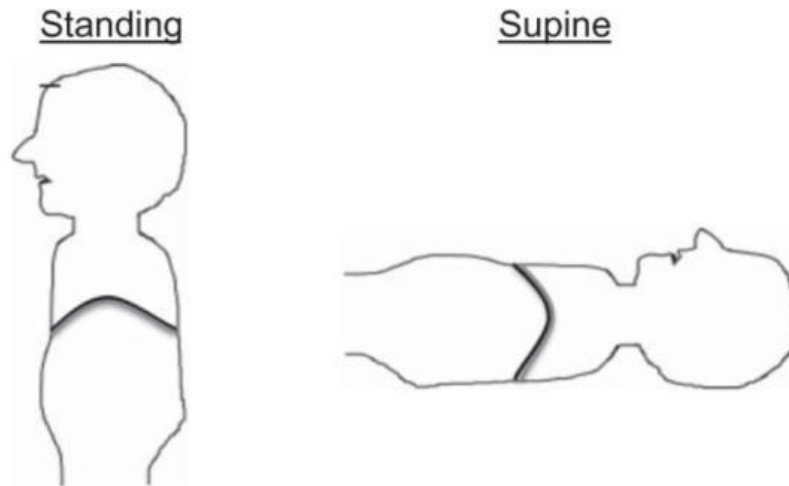
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Table 16.3 Terms Used to Describe Lung Volumes and Capacities

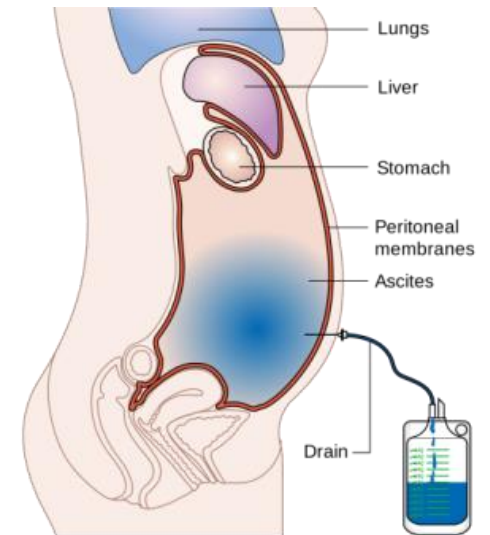
Term	Definition
<i>Lung Volumes</i>	The four nonoverlapping components of the total lung capacity
Tidal volume	The volume of gas inspired or expired in an unforced respiratory cycle
Inspiratory reserve volume	The maximum volume of gas that can be inspired during forced breathing in addition to tidal volume
Expiratory reserve volume	The maximum volume of gas that can be expired during forced breathing in addition to tidal volume
Residual volume	The volume of gas remaining in the lungs after a maximum expiration
<i>Lung Capacities</i>	Measurements that are the sum of two or more lung volumes
Total lung capacity	The total amount of gas in the lungs after a maximum inspiration
Vital capacity	The maximum amount of gas that can be expired after a maximum inspiration
Inspiratory capacity	The maximum amount of gas that can be inspired after a normal tidal expiration
Functional residual capacity	The amount of gas remaining in the lungs after a normal tidal expiration

Change in FRC

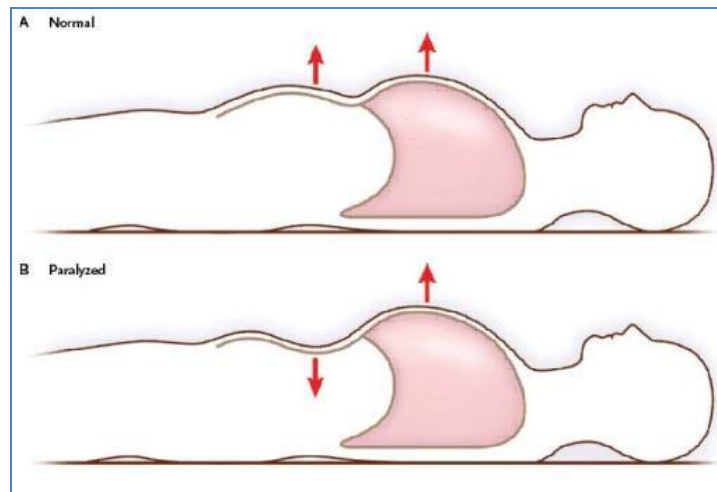
Normal



Ascites

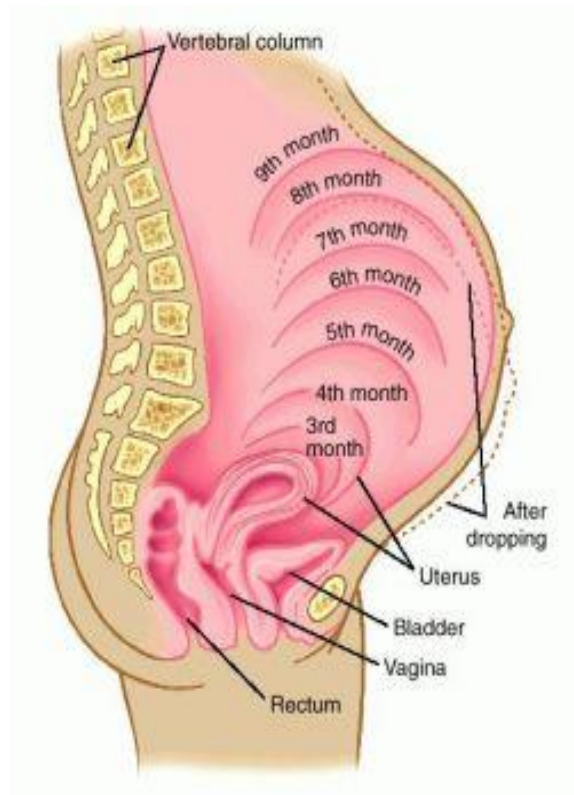


Paradoxical movement



Change in FRC

Pregnancy



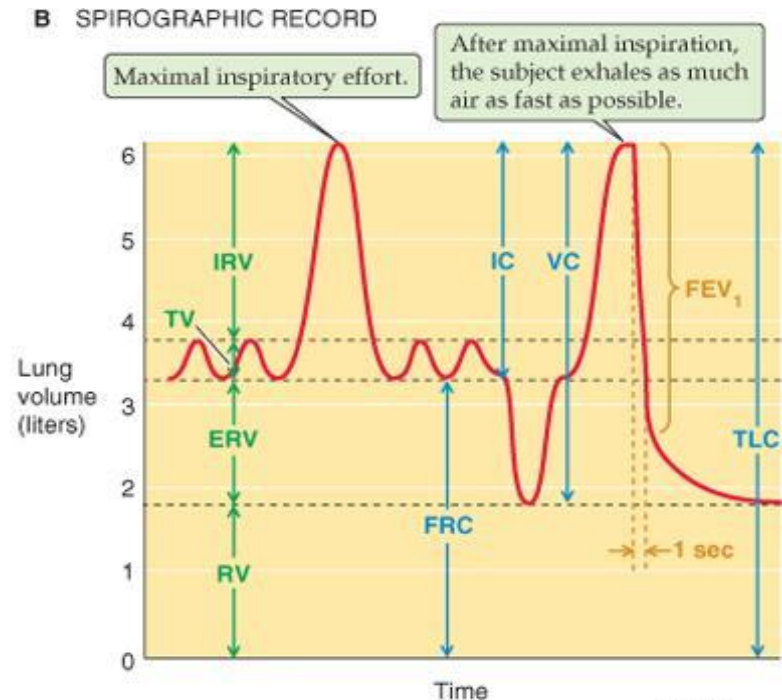
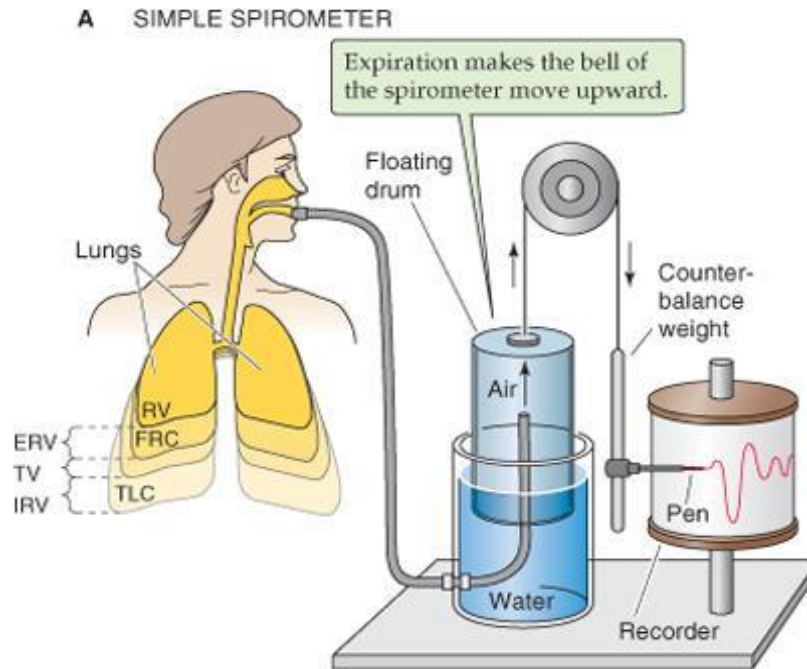
Obesity



Basic Concepts in Respiratory Physiology

1. Lung Volumes and Capacities
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Spirometer



Spirometer does not measure...

- Residual Volume

Volume and Capacities	Typical ranges (liters)
IRV = Inspiratory reserve volume	1.9–2.5
TV = Tidal volume	0.4–0.5
ERV = Expiratory reserve volume	1.1–1.5
RV = Residual volume	1.5–1.9
TLC = Total lung capacity	4.9–6.4
IC = Inspiratory capacity	2.3–3.0
FRC = Functional residual capacity	2.6–3.4
VC = Vital capacity	3.4–4.5

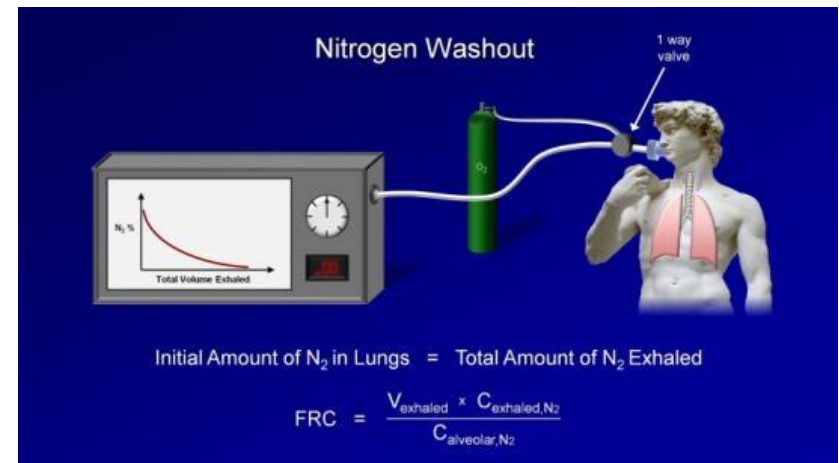
Determination of Residual Volume

1. Whole Body Plethysmography



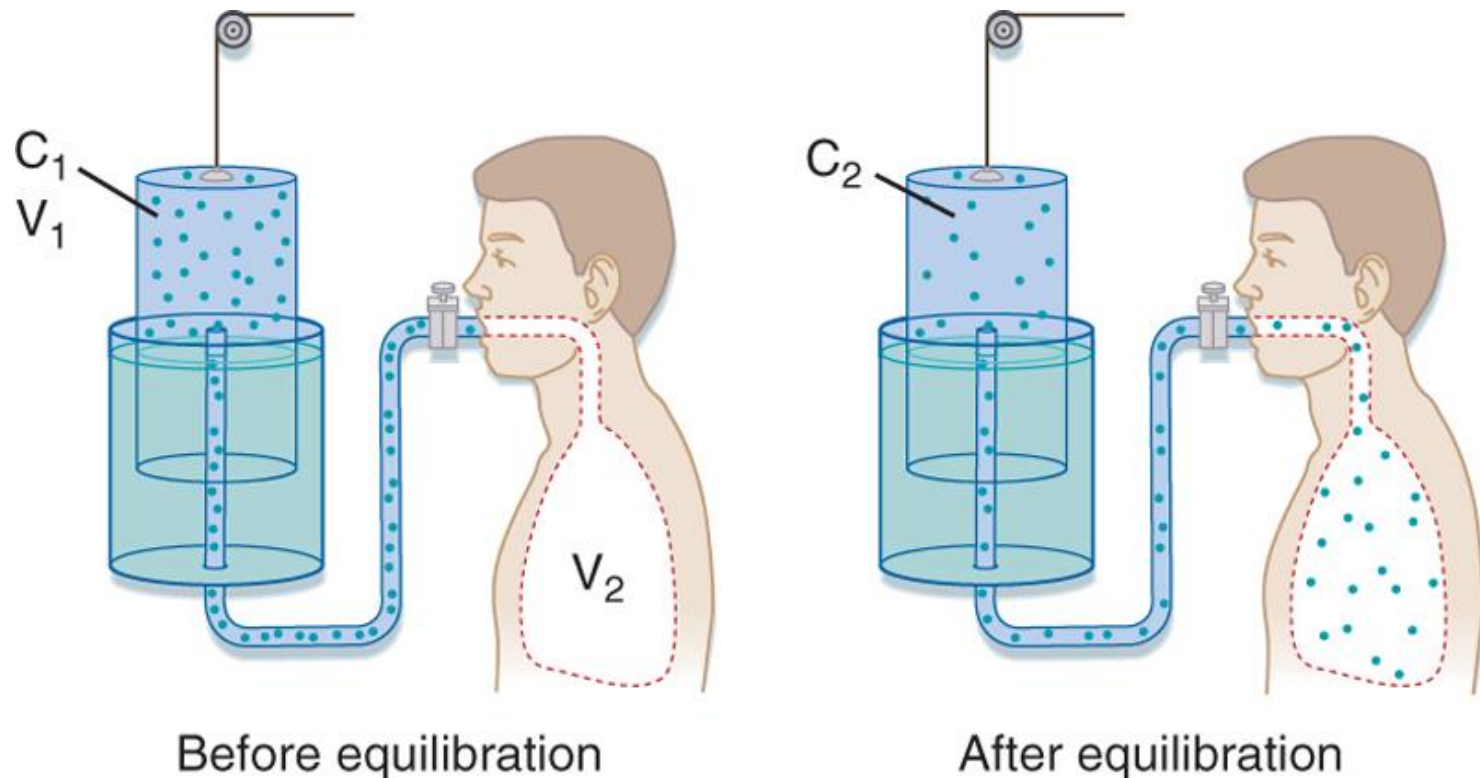
Boyle's law

- Gas Dilution Techniques
 2. Nitrogen Washout
 3. Helium Dilution



Helium Dilution Technique

An indirect method to measure the residual volume. Essential to know FRC which is important parameter for pulmonary diseases



$$C_1 \times V_1 = C_2 \times (V_1 + V_2)$$

Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.

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Calculations

$$C_1 \times V_1 = C_2 \times (V_1 + V_2)$$

$$V_1 + V_2 = \frac{(C_1 \times V_1)}{C_2}$$

$$V_2 = \frac{(C_1 \times V_1)}{C_2} - V_1$$

$$V_2 \text{ (FRC)} = \left(\frac{C_1}{C_2} - 1 \right) \times V_1$$

$$V_{\text{He}} = 1,950 \text{ ml}$$

$$[\text{He}]_{\text{initial}} = 10 \%$$

$$[\text{He}]_{\text{final}} = 3.9 \%$$

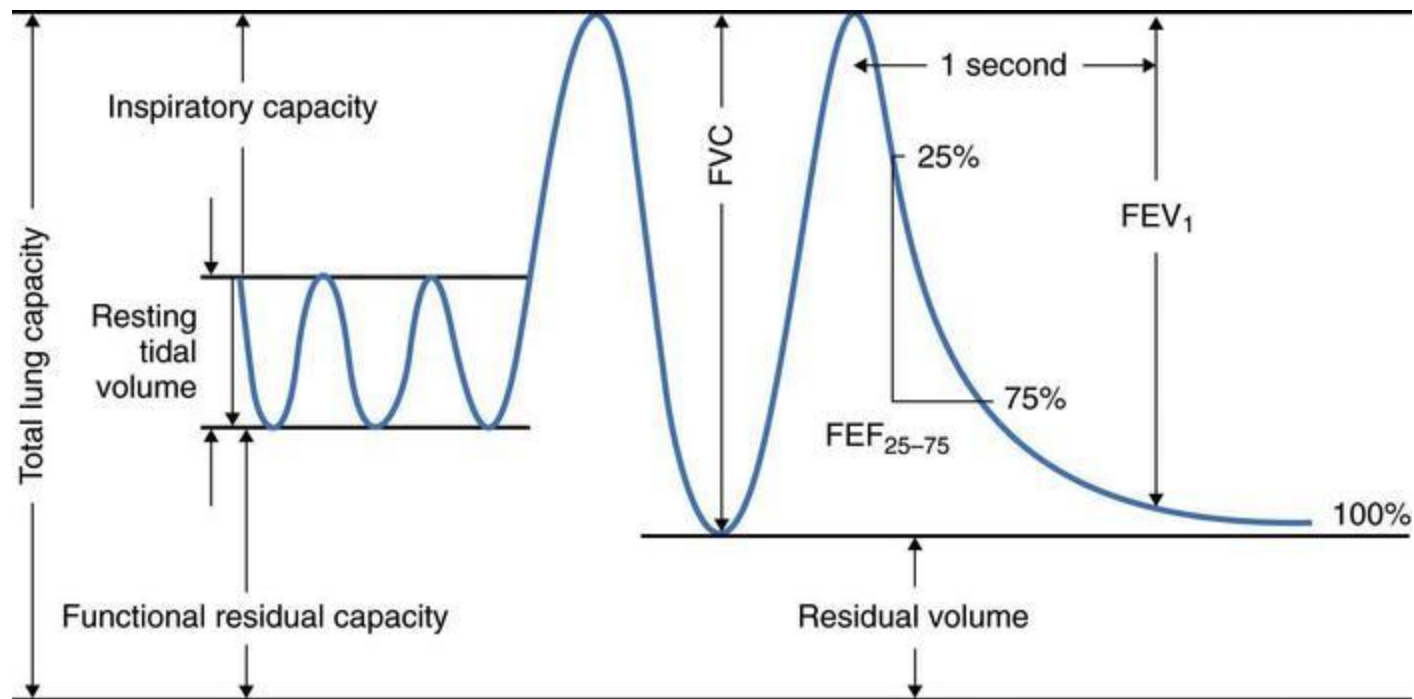
$$\text{ERV} = 1,200 \text{ ml}$$

$$\text{FRC} =$$

$$\text{RV} =$$

Forced Vital Capacity (FVC)

- Air exhaled as rapidly as possible after a maximum inhalation



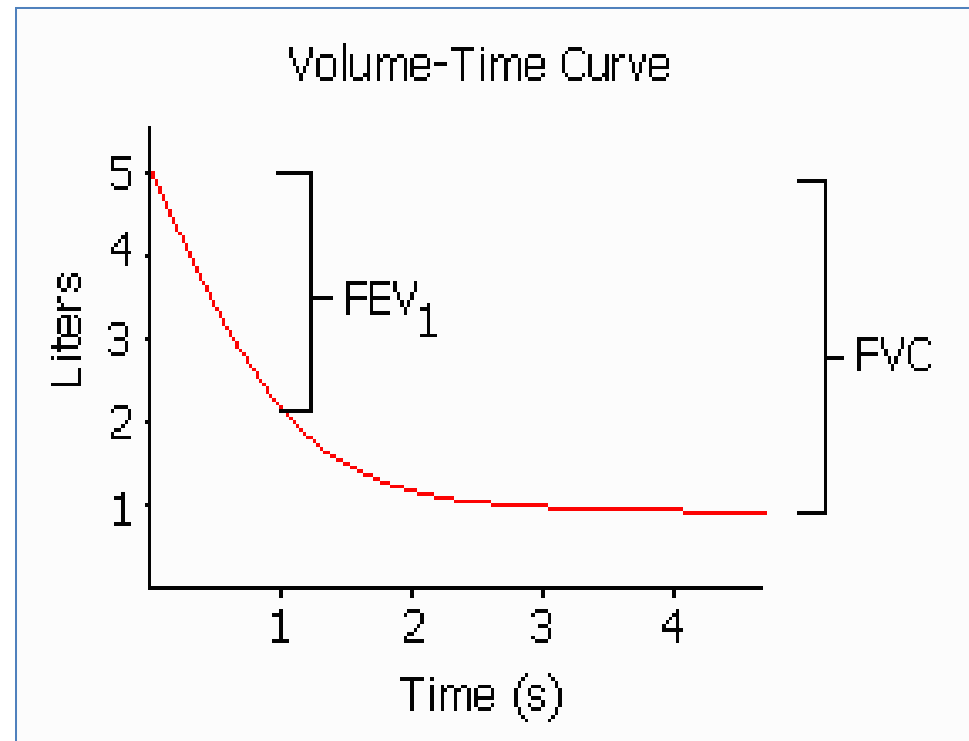
Brief Summary

FVC (Forced Vital Capacity) -- This is the maximum volume of gas that can be exhaled as forcefully and rapidly as possible after a maximal inhalation.

FEV₁ (Forced Expiratory Volume in 1st Second) -- This is the volume of air expired in the first second during maximal expiratory effort.

FEV₁/FVC -- This is the percentage of the vital capacity which is expired in the first second of maximal expiration. In healthy patients the FEV₁/FVC is usually around 70%.

Values are reported both as absolute values and as a percentage



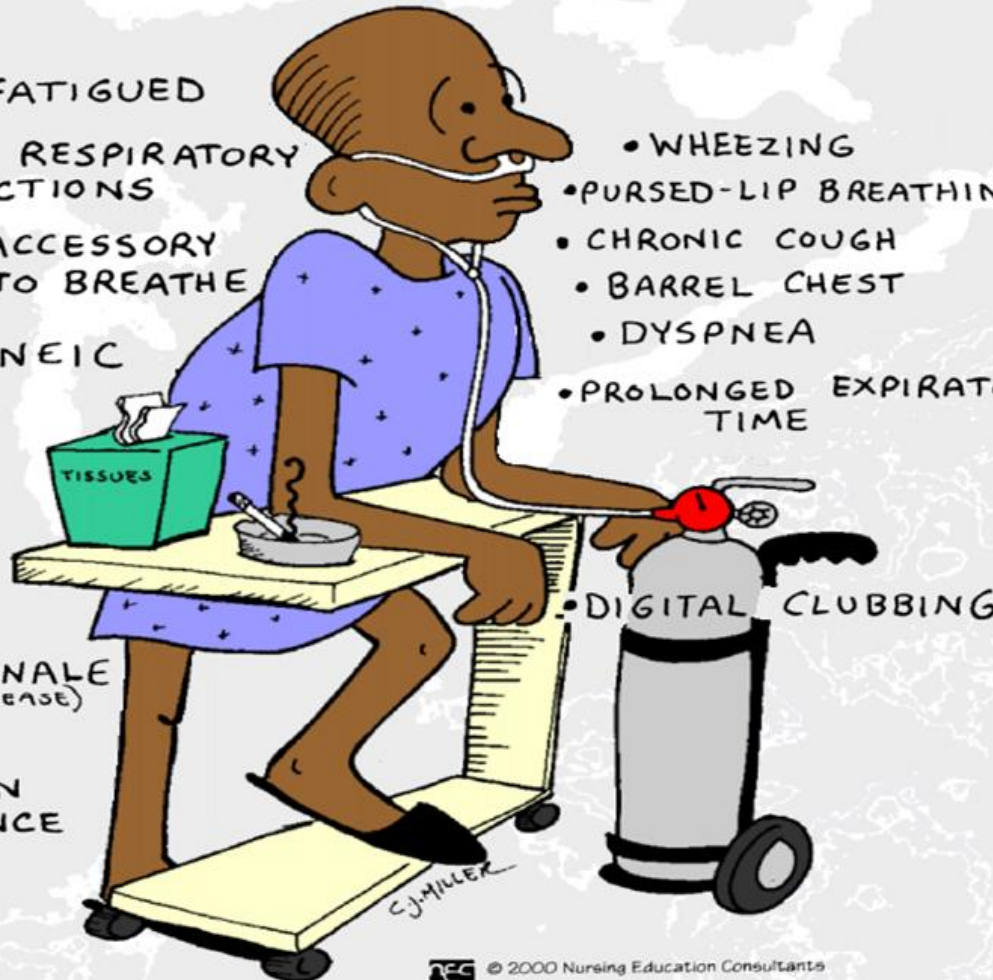
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COPD

CHRONIC AIRFLOW LIMITATION
"EMPHYSEMA AND CHRONIC BRONCHITIS"

- EASILY FATIGUED
- FREQUENT RESPIRATORY INFECTIONS
- USE OF ACCESSORY MUSCLES TO BREATHE
- ORTHOPNEIC
- COR PULMONALE (LATE IN DISEASE)
- THIN IN APPEARANCE
- WHEEZING
- PURSED-LIP BREATHING
- CHRONIC COUGH
- BARREL CHEST
- DYSPNEA
- PROLONGED EXPIRATORY TIME
- DIGITAL CLUBBING

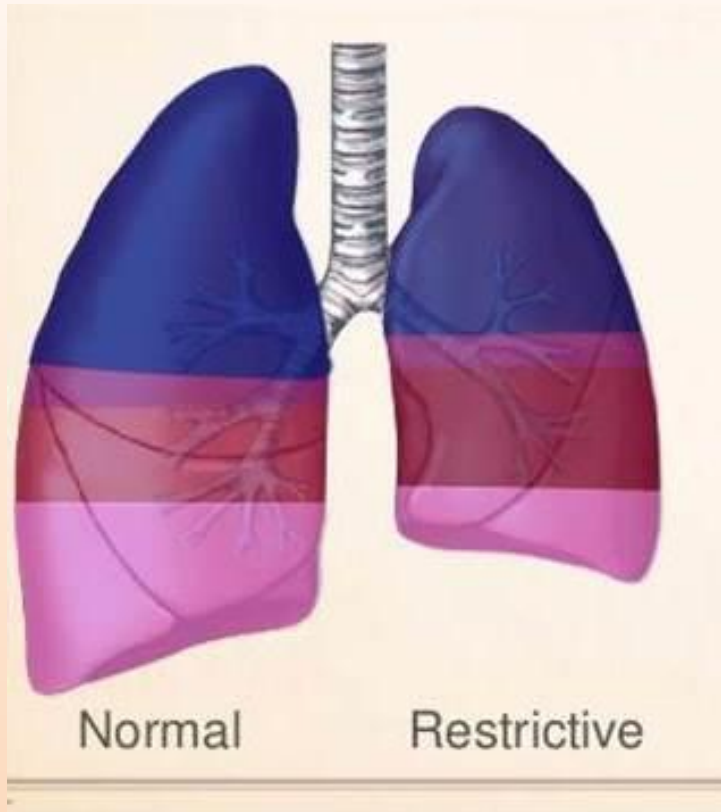


Obstructive Lung Disease



- Airway obstruction causes an *increase in resistance*. **Air trapping**
- Common obstructive diseases include asthma, bronchitis, and emphysema.
- $\downarrow$ FEV1, because of $\uparrow$ airway resistance
- FEV1/FVC decreased
- This is because it is very difficult for a person with an obstructive disease (eg. asthma) to exhale quickly due to the increase in airway resistance.
- RV $\uparrow$

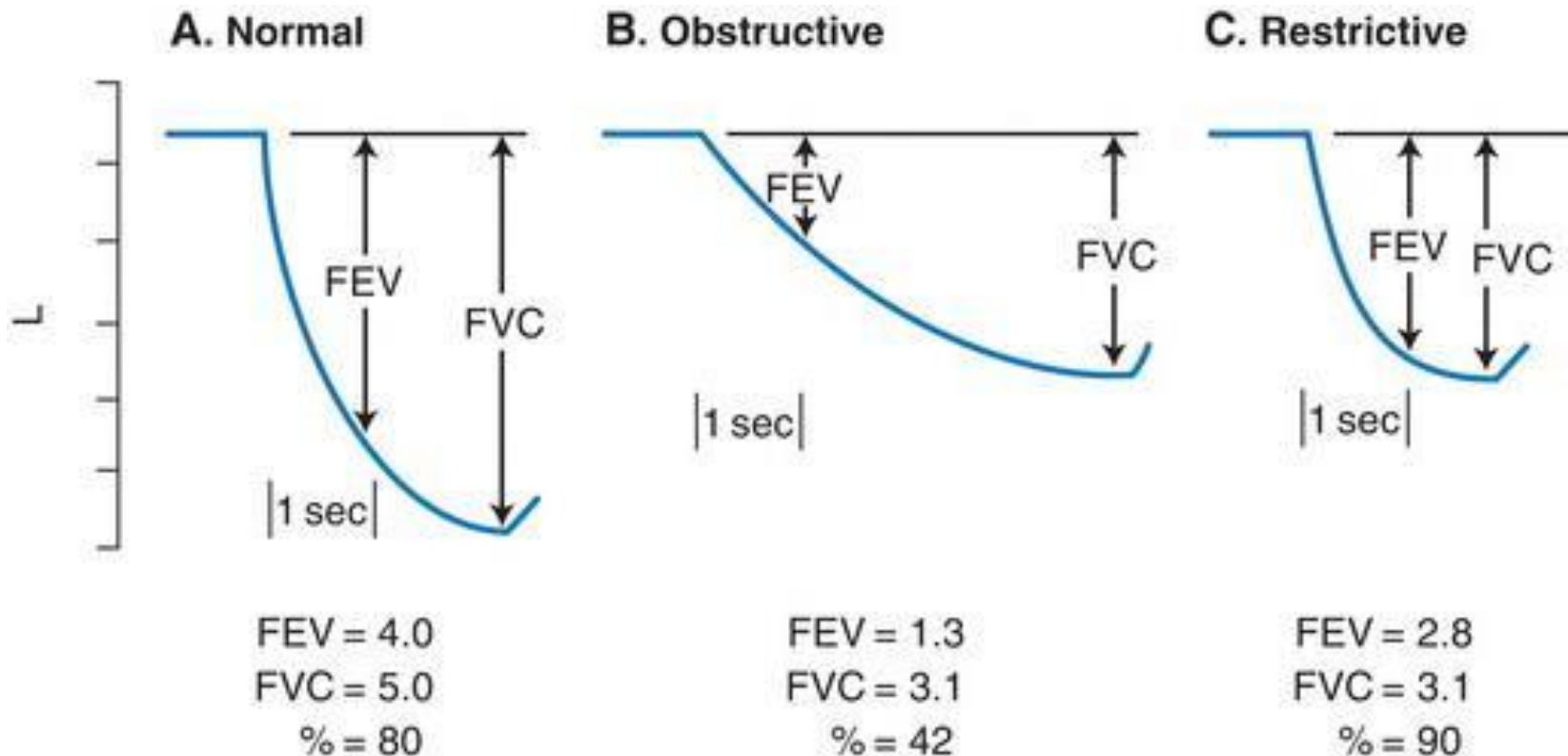
Restrictive Lung Disease



- The compliance of the lung is reduced, which increases the stiffness of the lung and limits expansion.
- Common causes of decreased lung compliance are pulmonary fibrosis, pneumonia and pulmonary edema.
- ↓ FVC.
- FEV1 is relatively large in comparison i.e. the FEV1/FVC ratio can be higher than normal.
- This is because it is easy for a person with a restricted lung to breathe out quickly, because of the high elastic recoil of the stiff lungs
- RV ↓



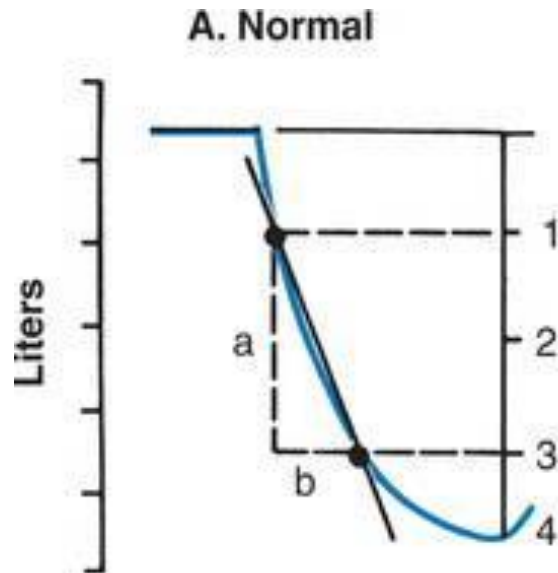
FEV1 and FVC in Lung Diseases



Physiology of Lung Diseases

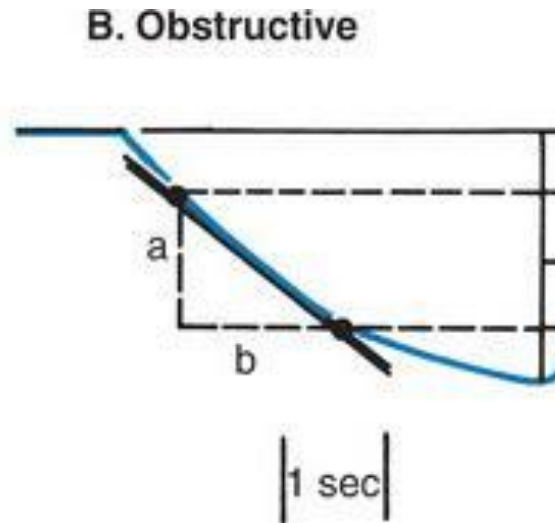
Disease	FEV ₁	FVC	FEV ₁ /FVC	FRC
COPD/Asthma	↓↓	↓	↓	↑
Fibrosis	↓	↓↓	↑ (or normal)	↓

Forced expiratory flow from a forced expiration

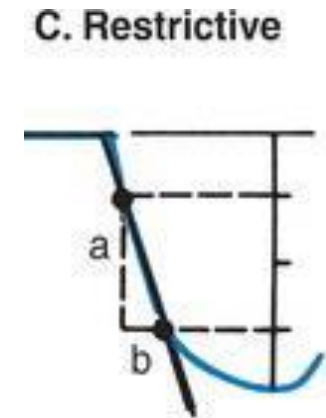


$$\text{FEF}_{25-75\%} = \frac{a}{b}$$

$$= 3.5 \text{ L/sec}$$



$$\text{FEF}_{25-75\%} = 1.4$$



$$\text{FEF}_{25-75\%} = 3.7$$

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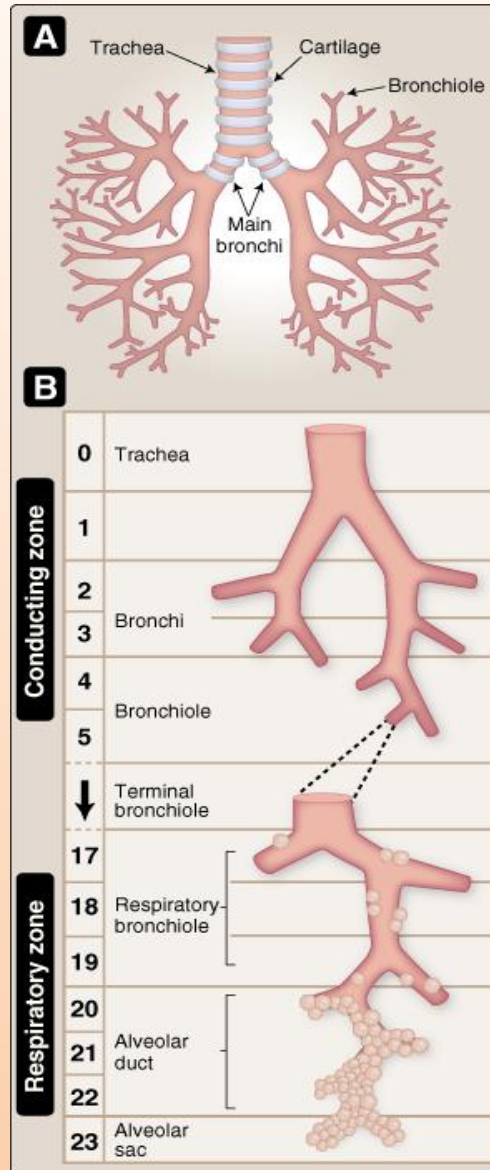
Dead Space

- The portion of each tidal volume that does not take part in gas exchange
- *Wasted Ventilation*

Total Dead Space =

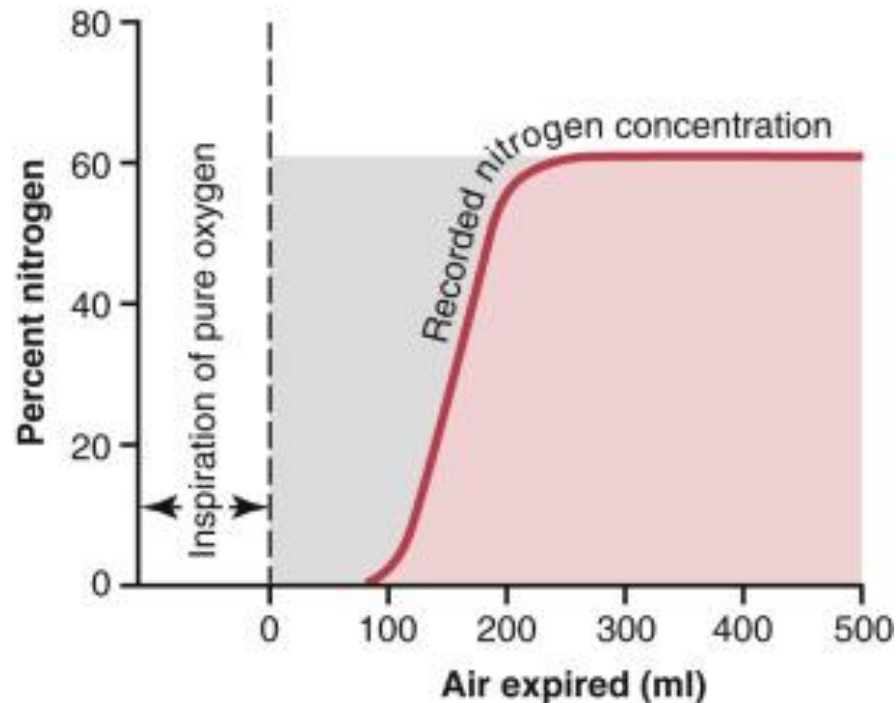
**Anatomic Dead Space
+
Alveolar Dead Space**

Anatomic Dead Space



Measurement of Anatomical Dead Space

Nitrogen Dilution - Fowler's Method

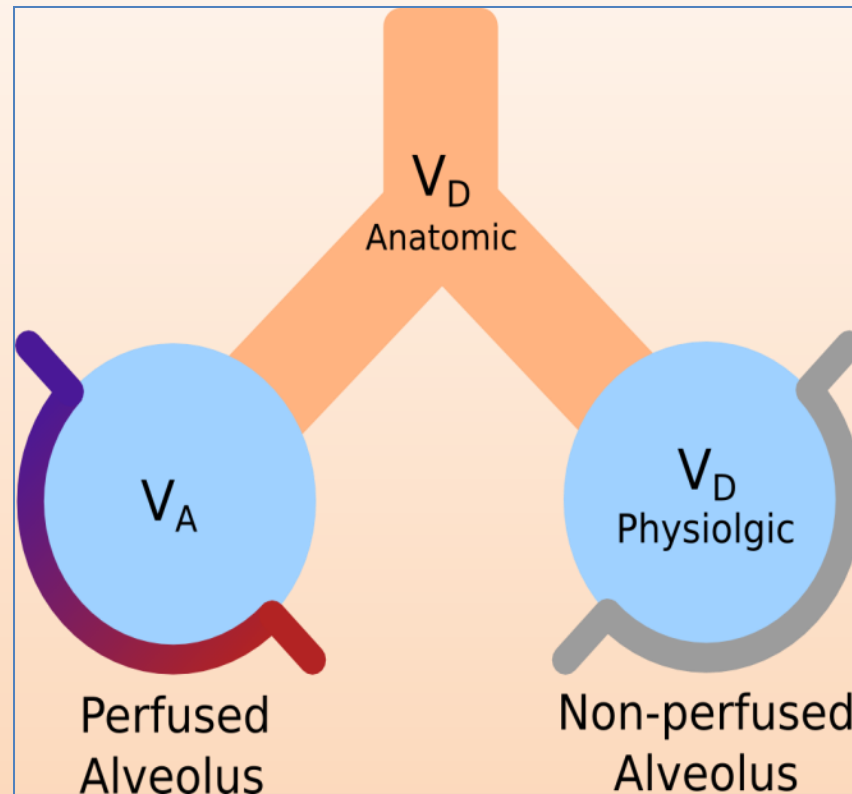


$$V_D = \frac{\text{Gray area} \times V_E}{(\text{Pink area} + \text{Gray area})}$$

where V_D is dead space air
and V_E is the total volume of
expired air.

A record of the changes in nitrogen concentration in the expired air after a single previous inspiration of pure oxygen.

Alveolar Dead Space



- Alveoli that are ventilated but not perfused, and where, as a result, no gas exchange can occur
- Apex is largest contributor

Measurement of Anatomical and Alveolar Dead Spaces

TV multiplied by the dilution of alveolar PCO_2 by dead space air, which does not participate in gas exchange and does not therefore contribute CO_2 to expired air.

Bohr-Enghoff Method

$$\frac{V_D}{V_T} = \frac{P_{ACO_2} - P_{ECO_2}}{P_{ACO_2}}$$

V_D = physiologic dead space (mL)

V_T = tidal volume (mL)

P_{ACO_2} = PCO_2 of alveolar gas (mm Hg) = P_{aCO_2} of arterial blood

P_{ECO_2} = PCO_2 of expired air (mm Hg)

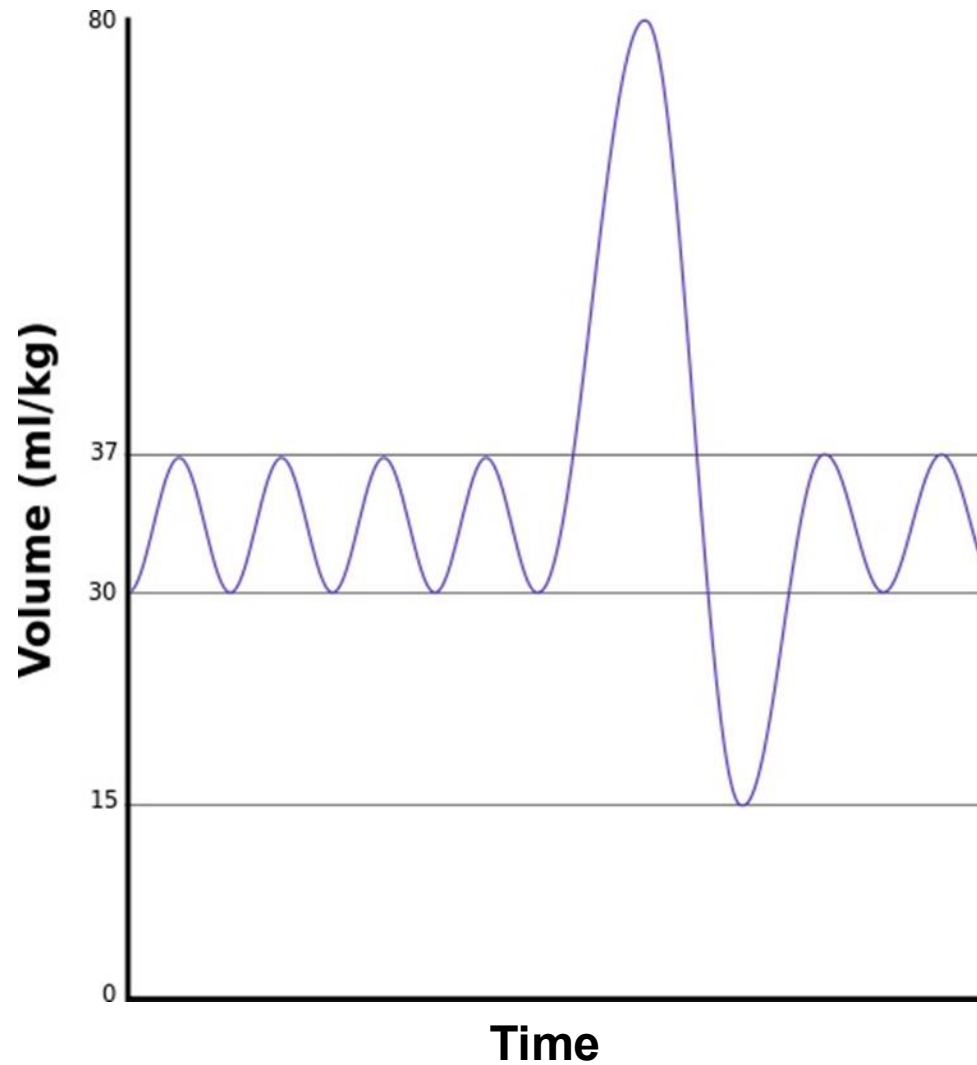
Summary

Most of the lung volumes and capacities can be directly measured via spirometry.

The residual volume (RV) can serve a valuable indicator of pulmonary disease and is derived indirectly.

Restrictive and obstructive lung disease can be distinguished by duration and volume of expiratory air release.

Quick Quiz



Learning Objectives

1. Compare and contrast the relationships between lung volumes and capacities.
2. Discuss spirometry and its limitations in measurements of lung volumes and capacities.
3. Outline methods to determine residual volume and functional residual capacity.
4. Extrapolate lung volumes and capacities with respect to obstructive and restrictive lung diseases.
5. Demarcate regions of the lung that do not participate in gas exchange and learn methods to measure dead space.

More sample questions for practicing

1. <https://exchange.scholarrx.com/brick/pulmonary-function-tests>
2. <https://exchange.scholarrx.com/brick/obstructive-and-restrictive-lung-diseases-foundations-and-frameworks>
3. <https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=258031997&bookid=3310>
4. <https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=49980261&bookid=811>
5. <https://casefiles-mhmedical-com.nyit.idm.oclc.org/CaseContent.aspx?gbosID=218980&gbosContainerID=68&viewByNumber=false&groupid=0#104658253>

References

- **Guyton and Hall's Textbook of Medical Physiology**, 14th and 15th Eds. Chapters 38, 40, 43 (as applicable to lecture extent)
 - Copy & paste should work; direct-click may not work sometimes
 - <https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20170004883>
 - <https://www.clinicalkey-com.nyit.idm.oclc.org/#!/browse/book/3-s2.0-C20220008424>
- **West's Pulmonary Pathophysiology: The essentials**, 10th Ed.; Chapters 1, 4, 5 (as applicable to lecture extent)
 - Copy & paste should work; direct-click may not work sometimes
 - <https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.wwhealthlibrary.com/book.aspx?bookid=3076>
- **ScholarRx Brick(s) and link(s):**
 - Pulmonary Function Tests
 - <https://exchange.scholarrx.com/brick/pulmonary-function-tests>
 - Obstructive and Restrictive Lung Diseases: Foundations and Frameworks
 - <https://exchange.scholarrx.com/brick/obstructive-and-restrictive-lung-diseases-foundations-and-frameworks>



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XREL4R](https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R)